# Save The Cat? Assessing the Ability to Predict Global Box Office Based on Adherence to Story Structure

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# Abstract

Given an increasingly risk-averse entertainment industry, research into the financial impact of story structure is highly needed. While there has been much literary scholarship about story structure, very little research has been produced using mathematical modeling to estimate financial output. This paper seeks to determine if adherence to the beat structure from Blake Snyder's *Save The Cat! (2005)* can be used to predict global box office. Using two Ordinary Least Squares models and a Random Forest regression, the study looks at a sample of 51 films evaluated for beat adherence by Anthropic's LLM Claude 3.7 Sonnet. The models fail to find a statistically significant relationship between deviations from Snyder's beats and financial output. Outliers rooted in existing intellectual properties dominated the box office, showing that brand recognition appears to be a much greater factor than story structure. We conclude that story structure alone cannot be used to predict global box office.

# Acknowledgements

First and foremost, I would like to thank my parents Lindsey and Scott Toban for their constant support.

I would also like to thank my advisor, Prof. Tracey Scott Wilson, for her invaluable wisdom about storytelling.

For his leadership, I would like to acknowledge Joe Anoa'i.

Finally, I would like to express gratitude to Kaveh Daneshmand of Prague Film School for the most useful creative feedback I have ever received, and the statement that spawned this research: "Your story was technically perfect. I felt nothing."

## 1. Introduction

### 1.1 Seeking a Formula

The entertainment industry has undergone tremendous changes in recent years—according to the Writer's Guild of America (WGA), the number of jobs in television writing fell 42% in the 2023-2024 season from the year prior. Parrot Analytics reported that in Q4 of 2024, existing intellectual properties accounted for 34% and 38% of customer acquisition and retention respectively across film and television. These both suggest increasingly risk avoidant tendencies by studios and networks, largely depending upon audience familiarity with established brands or the reliability of established writers.

This raises a number of questions for Hollywood: Are audience familiarity and a smaller pool of writers the only way to reduce the risk of box office failure? What if the creative elements, such as story structure, could be optimized to minimize risk as well? To offer one answer to this question, this paper examines the predictive power of measurable adherence to a well-known story structure—Blake Snyder's *Save The Cat!*—on global box office. By testing this mathematically, we can perhaps provide evidence that the studios do not need to be so reliant on these other factors if the story structure is reliable.

To test this hypothesis, I assembled a dataset of 51 films by extracting story maps from each individual screenplay using a Large Language Model (LLM). From this, I measured each film's deviations from Snyder's outline in *Save The Cat!* and then tested these against inflation-adjusted global box office with three mathematical models. All three of my models showed little to no statistically significant relationship between beat

adherence and box office revenue. Before outlining the methodology, the following subsection explains the current state of the field in storytelling research, the mathematical gaps, and the novelty of this study.

### 1.2 Literature Review

Non-mathematical theories on storytelling have existed for centuries, from Aristotle all the way to Blake Snyder. Many books have been written on screenplay structure, most famously *Story* (1997) by Robert McKee, whose work is frequently assigned in introductory screenwriting courses. While McKee offered insights on scene structure, pacing, and rough estimates on when major developments should occur, his philosophy was far from scientific. Paul Gulino's *Screenwriting: The Sequence Approach (2004)* offered a somewhat more defined approach, dictating what should happen in eight "sequences," each ten to fifteen minutes, with clearer prescriptions for what changes must occur in each part. However, Snyder's *Save The Cat!* is by far the most scientific, prescribing exactly fifteen clearly defined story beats with suggested start and end pages for each one. The book also includes examples of each beat in successful contemporary films as evidence of his theory. What all three of these writers have in common is that their evidence is completely anecdotal—what do mathematicians have to say about story structure?

Currently, the answer is not much. A 2007 study by Eliashberg, Hui, and Zhang utilized a Bag-CART model to demonstrate a relationship between viewer summaries immediately after seeing a film and returns on investment. This is the only example I have found of scholars measuring story against financial output, and it does not rely

directly upon the structure of any screenplays. Otherwise, most studies of film economics look at more easily measured factors like star power. For example, another 2007 study by Anita Elberse looked at the effects of star casting announcements on the financial success of films. This area of research for financial forecasting of films is far more common and established. Thus, my survey of existing literature suggests that there is a wide gap in the canon of formal mathematical research on cinema, specifically as it relates to story structure. My paper aims to provide a beginning.

## 2. Setting/Data

Shown in **Table 1** are the start and end pages of all fifteen of Blake Snyder's beats in a theoretical 110-page screenplay, sourced directly from *Save The Cat!*

***Table 1: Save the Cat! Beat Sheet Page Ranges***

| Beat | Start Page | End Page |
|---|---:|---:|
| Opening Image | 1 | 1 |
| Theme Stated | 5 | 5 |
| Set-Up | 1 | 10 |
| Catalyst | 12 | 12 |
| Debate | 12 | 25 |
| Break Into Two | 25 | 30 |
| B Story | 30 | 35 |
| Fun and Games | 30 | 55 |
| Midpoint | 55 | 55 |

| Beat | Start Page | End Page |
|---|---|---|
| Bad Guys Close In | 55 | 75 |
| All Is Lost | 75 | 80 |
| Dark Night of the Soul | 80 | 85 |
| Break Into Three | 85 | 90 |
| Finale | 90 | 99 |
| Final Image | 99 | 110 |

***Table 1: Save the Cat! Beat Sheet Page Ranges***

In order to examine the impact of story structure on global box office, we first must compile a dataset of start points for each of these beats in real screenplays. I achieved this using Anthropic's Large Language Model (LLM) Claude 3.7 Sonnet by uploading individual screenplays after the following prompt (workshopped from a prompt generated by Claude):

> "You are a professional script analyst with deep expertise in story structure. I will upload a screenplay in standard format. Your task is to identify the exact start and end pages for each of the 15 canonical beats in Blake Snyder's Save the Cat beat sheet. Use only the explicit content of the screenplay - do not infer from generic page formulas. Your output should be a CSV table with the following columns: Beat,Page_Start,Page_End,Description Required Beats (must appear in order and without omission): 1. Opening Image 2. Theme Stated 3. Set-Up 4. Catalyst 5. Debate 6. Break Into Two 7. B Story 8. Fun and Games 9. Midpoint 10. Bad Guys Close In 11. All ls Lost 12. Dark Night of the Soul 13. Break Into Three 14. Finale 15. Final Image Rules:* Do not guess page numbers from general formulas (e.g.,"Catalyst= page 12"). * Identify beats based on actual narrative transitions: character decisions, tone shifts, location/context changes, or key dialogue. * Each beat must occupy a distinct, continuous page range - no overlaps. * If a beat is ambiguous, select the most defensible start and end pages based on dramatic structure and script content. * Do not merge or omit any beats. After the upload, begin by reading all pages and segmenting the screenplay sequentially. Then return only the CSV output, nothing else."

With this method, I manually compiled the dataset from 51 films. By asking the LLM to justify its decision for every beat, I allowed myself to perform a human sanity check on each script to make sure the data was consistent with storytelling logic. Most importantly, I was able to confirm that the LLM was actually reading the script rather than guessing based on external resources.

Using these numbers, I converted the start pages to percentages of each script starting with the figures in **Table 1.** This allowed me find percent deviations for each beat by subtracting Snyder's start percentage from the start percentage in each datapoint. With the individual deviations for every single beat, I also calculated a mean deviation for each film to be used in **Model 1.** Global box office numbers were sourced directly from IMDB Pro, and then manually adjusted for inflation.

## 3. Methodology

### 3.1. Model 1 (OLS)

#### 3.1.1 Equation

The first model is an Ordinary Least Squares (OLS) regression that estimates global box office as a linear function of each film's average deviation from Snyder's beat sheet. The equation if given by:

$$Y_i = \beta_0 + \beta_1 \cdot \text{AverageDeviation}_i + \varepsilon_i$$

This model fits $Y_i$, the global box office for each film *i,* against the average deviation across all beats in every film. $\beta_1$ represents the coefficient for the deviation and is the only dependent variable in the model.

### 3.1.2 Assumptions

We follow the standard assumptions for a univariate OLS model:

1. Linearity
2. Independence of Errors
3. Homoscedasticity
4. Normality of Errors
5. Exogeneity

### 3.1.3 Caveats

While this model will provide summary statistics that are easily interpretable, it relies only upon average deviation; it does not account for the varying impacts of each individual beat. It also can be easily affected by outliers, which would cause massive shifts in the regression. The individual assumptions also must be verified in the results of the model—if the model shows signs of heteroscedasticity or if the errors are not normal, the model will not hold predictive power but will still directly address the research question.

## 3.2. Model 2 (OLS)

### 3.2.1 Equation

The second model is also an OLS regression; however this one is multivariate. Here we regress global box office against the deviations from all fifteen individual beats in every film. The equation is given by:

$$Y_i = \beta_0 + \sum_{k=1}^{15} \beta_k \cdot X_{ik} + \varepsilon_i$$

This function fits $Y_i$, the global box office for each film *i,* against the individual deviations from each beat *j* in every film. $X_{ji}$ denotes a matrix of all these deviations for the fifteen beats in each film, where $\beta_j$ represents the coefficient for the deviations from each beat.

### 3.2.2 Assumptions

We follow the same five assumptions as Model 1, however we must add a sixth: no perfect multicollinearity.

### 3.2.3 Caveats

The same caveats apply to this OLS model as the univariate one, however now we are accounting for the impacts of individual beats. With a sample size of fifty-one and a model with fifteen predictors, this model is very likely to overfit. The small sample size also likely hinders the model's ability to identify small effects. While the measured magnitude of each effect produced by this model might be unreliable, it is still useful for identifying the relationship at the core of the research question.

### 3.3. Model 3 (Random Forest)

#### 3.3.1 Equation

Similarly to the previous model, we are once again looking at global box office as a function of the individual beat deviations. Now however, we are accounting for the possibility of nonlinear relationships using a Random Forest Model. While a specific function for these models cannot be given, the basic concept of the model can be explained by:

$$\widehat{Y_i} = \frac{1}{T}\sum_{t=1}^{T} f_t(X_i)$$

is an estimation of global box office for film *i* as an average of $f_t(X_i)$ across all trees *t*. In a random forest model, *T* trees are trained different subsets of the data to generate individual predictions. I have set the parameter T=500, meaning the random forest model averages from five hundred trees.

#### 3.3.2 Assumptions

The benefit of this model is that it does not assume linearity, homoscedasticity, or normality of errors. As we will see in the results, these assumptions largely do not hold for either OLS model. The key assumptions are simply independence of observations and the existence of any kind of informative pattern.

### 3.3.3 Caveats

The nature of this model makes it more difficult to interpret, as the summary statistics are not as concrete as the OLS models. There will not be coefficients for each beat, but this model does fill in the most major gap from the linear models by accounting for nonlinear relationships between each beat and global box office. Also, because Random Forests are not parametric like the OLS models, they are unlikely to be applicable outside of the sample.

## 4. Results

### 4.1 Model 1

### 4.1.1 Summary Statistics

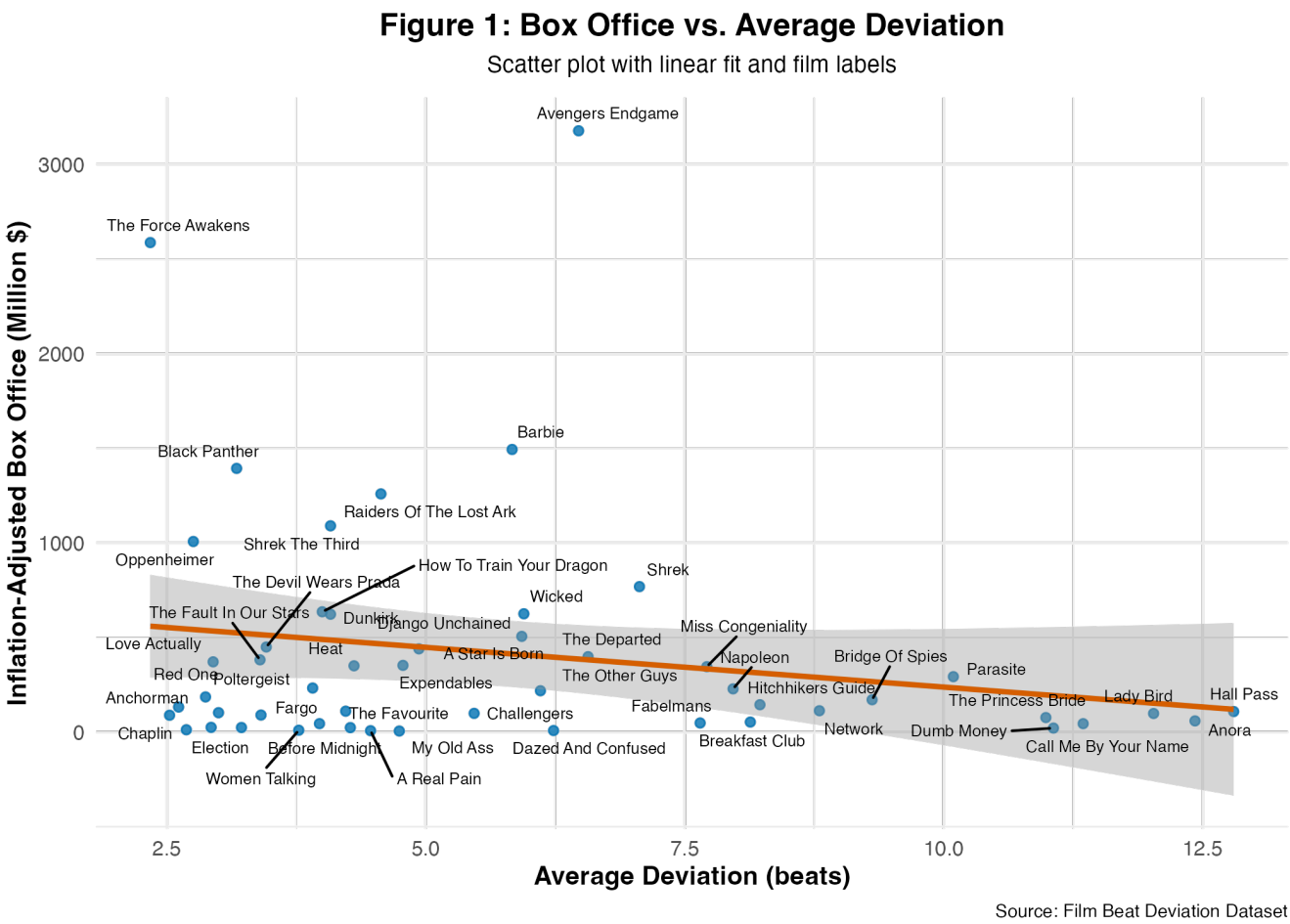
***Table 2: OLS Model 1 - Average Deviation Predicting Box Office***

| Term | Estimate | Std. Error | t value | p value | Conf. Low | Conf. High |
|---|---|---|---|---|---|---|
| (Intercept) | 658.107 | 193.970 | 3.393 | 0.001 | 268.311 | 1,047.904 |
| average_deviation | -42.011 | 29.911 | -1.405 | 0.166 | -102.119 | 18.098 |

The summary statistics for the first model include a p-value of 0.166 for the coefficient, meaning that with the standard α=.05, the results of this regression are not statistically significant. A t-value of -1.405 is also not statistically significant at this threshold. The coefficient of -42.011 indicates a negative relationship between average deviation and box office, with a increase of 1% in average deviation corresponding to a loss of roughly $42 million. While this may be true of the coefficient, the confidence interval includes zero, meaning even the negativity is not statistically significant.

With all of these figures in mind, there is no statistically significant evidence that average deviation has a linear relationship with global box office. By looking at the scatterplots themselves, we can see more clearly what assumptions may have been violated and what external variables may be confounding the predictive power of these story deviations alone.

### 4.1.2 Visual Analysis

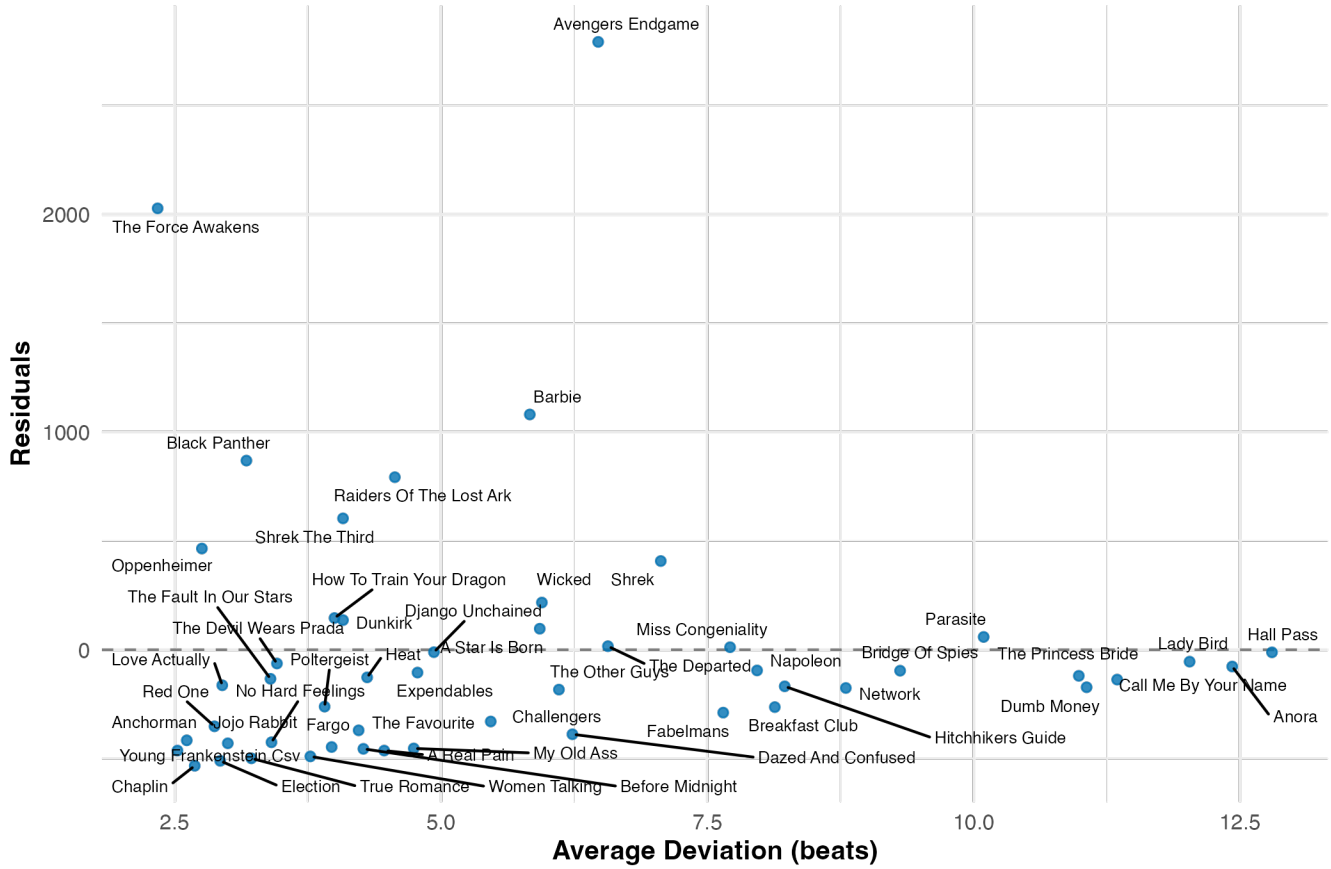



The gray region around the fit line is a confidence band, which shows the range of values within 95% confidence for mean predicted box office at each value for average deviation. The proportion of films inside the band becomes greater as the average

deviation becomes larger, indicating that the model forms more stable predictions for films that deviate further from Snyder's structure. The four largest outliers are *Avengers: Endgame, Star Wars: The Force Awakens, Barbie*, and *Black Panther* all of which belong to an established brand or franchise. This suggests that audience familiarity with existing intellectual property has a confounding effect on the model, and perhaps accounts more for global box office than story structure—at the very least, it seems to limit the explanatory power of average deviation in this setup.

The residuals plot makes this more apparent, and also reveals more clearly which assumptions have been violated:

**Figure 2: Residuals of Model 1: Box Office vs. Average Deviation**




Model residuals labeled by film

Here we see residuals greatly decreasing in magnitude as films deviate further from Snyder's structure, forming a rightward funnel in the plot. This is evidence that the assumption of homoscedasticity has been violated, as there is clear evidence of decreasing variance along the plot. The presence of these massive outliers also indicates the distribution of residuals is non normal, meaning that the assumption of normality of errors has been violated too. By violating these two assumptions, the plot suggests that average deviations from Snyder's beat sheet do not have a linear relationship with global box office. However, because the residuals follow no clear pattern, we cannot identify a nonlinear relationship using this model alone.

### 4.2 Model 2

#### 4.2.1 Summary Statistics

***Table 3: OLS Model 2 - All Beat Deviations Predicting Box Office***

| Term | Estimate | Std. Error | t value | p value | Conf. Low | Conf. High |
|---|---|---|---|---|---|---|
| (Intercept) | 413.879 | 475.101 | 0.871 | 0.390 | -550.627 | 1,378.385 |
| deviation_opening_image | 236.832 | 386.872 | 0.612 | 0.544 | -548.560 | 1,022.224 |
| deviation_theme_stated | -8.871 | 31.084 | -0.285 | 0.777 | -71.974 | 54.233 |
| deviation_set_up | 57.558 | 98.262 | 0.586 | 0.562 | -141.924 | 257.040 |
| deviation_catalyst | 16.718 | 42.506 | 0.393 | 0.696 | -69.574 | 103.009 |
| deviation_debate | 14.150 | 52.858 | 0.268 | 0.791 | -93.158 | 121.458 |
| deviation_break_into_two | -27.586 | 50.904 | -0.542 | 0.591 | -130.926 | 75.754 |
| deviation_b_story | 23.612 | 37.399 | 0.631 | 0.532 | -52.312 | 99.537 |
| deviation_fun_and_games | -33.756 | 37.420 | -0.902 | 0.373 | -109.722 | 42.210 |

| | | | | | | |
|---|---|---|---|---|---|---|
| deviation_midpoint | 14.388 | 35.687 | 0.403 | 0.689 | -58.061 | 86.837 |
| deviation_bad_guys_close_in | -38.392 | 39.166 | -0.980 | 0.334 | -117.903 | 41.120 |
| deviation_all_is_lost | 21.053 | 48.266 | 0.436 | 0.665 | -76.932 | 119.039 |
| deviation_dark_night_of_the_soul | -12.182 | 47.003 | -0.259 | 0.797 | -107.603 | 83.240 |
| deviation_break_into_three | -21.873 | 48.383 | -0.452 | 0.654 | -120.095 | 76.349 |
| **Term** | **Estimate** | **Std. Error** | **t value** | **p value** | **Conf. Low** | **Conf. High** |
| deviation_finale | 32.068 | 37.674 | 0.851 | 0.400 | -44.415 | 108.552 |
| deviation_final_image | 22.676 | 31.443 | 0.721 | 0.476 | -41.157 | 86.509 |

***Table 3: OLS Model 2 - All Beat Deviations Predicting Box Office***

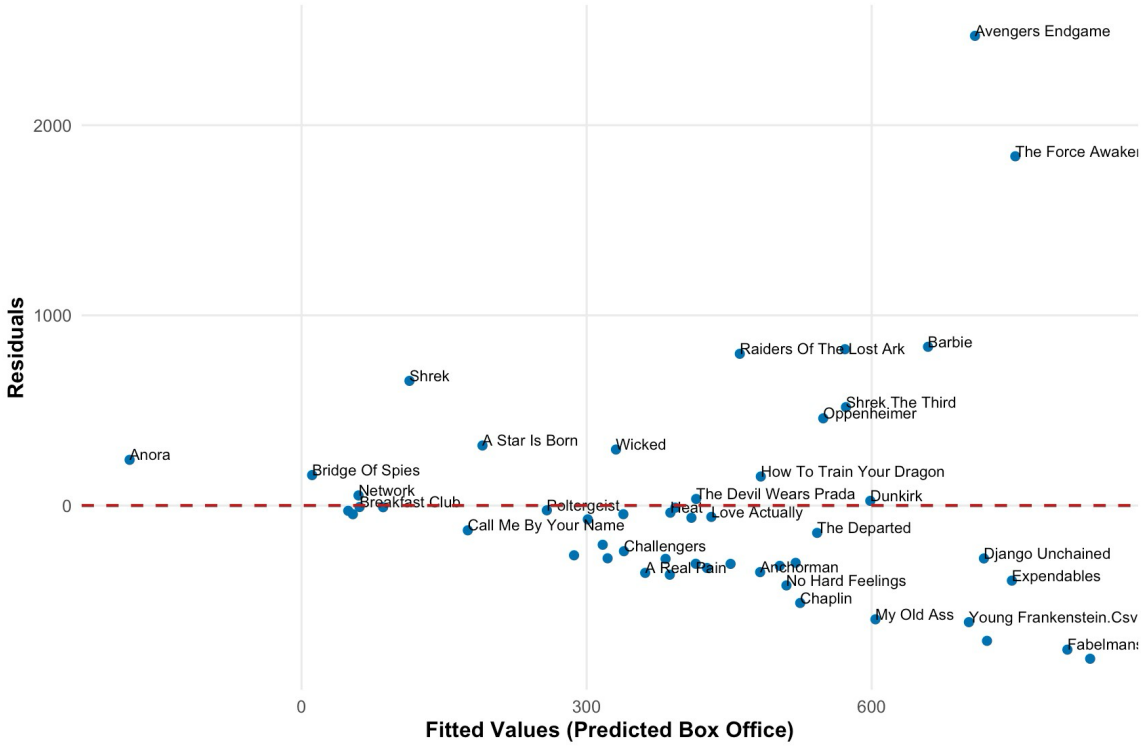
Of all the p-values for the fifteen beats, the lowest is 0.334. This means that not a single beat has a statistically significant coefficient. Also, every single 95% confidence interval includes zero, meaning there is no statistically reliable evidence of directionality for any of the beats either.

From a purely observational standpoint, there is no clear clustering within the table either. Beats in similar parts of the script show wildly different coefficients, such as the negative coefficient for “Break into Two” followed by the positive coefficient for “B Story.” This makes it harder to identify any kind of signal based on story structure from this model. Also, “Opening Image” has an abnormally large coefficient of $236.832 million, which is far outside the range of the entire rest of the beats, suggesting that the model is failing to stabilize for the noise in the dataset.

Given the failure of these summary statistics to identify a pattern, it is useful to once again look at a residuals plot in order to identify assumptions that have been violated and factors contributing to the model's instability.

### 4.2.2 Visual Analysis

**Figure 3: Residuals vs. Fitted Values (Model 2)**




This time, the residuals plot shows the deviations from the fitted values that the model predicted for box office. The plot clearly shows increasing variance for higher predicted values, suggesting that the model become less stable for higher grossing films. This is once again evidence of heteroscedasticity, violating the assumption of homoscedacticity. The presence of these large outliers on the residuals plot once again suggests that errors are not normally distributed, violating that assumption as well. These

violations corroborate the summary statistics' revelation that there is no evidence of a linear relationship.

This model also underestimated the same three largest outliers as the first: *Avengers: Endgame, Star Wars: The Force Awakens,* and *Barbie.* Even when story structure deviations are considered individually for each beat, the same films seem to show evidence that the explanatory power of this model is confounded by intellectual property.

***4.3 Model 3***

***Table 4: Random Forest Feature Importance***

| **Beat** | **Importance (%IncMSE)** | **Importance (%)** |
|---|---:|---:|
| deviation_all_is_lost | 4.241 | 33.82 |
| deviation_bad_guys_close_in | 3.523 | 28.10 |
| deviation_dark_night_of_the_soul | 2.385 | 19.02 |
| deviation_final_image | 1.062 | 8.47 |
| deviation_midpoint | 0.956 | 7.62 |
| deviation_break_into_two | 0.372 | 2.97 |
| deviation_opening_image | -3.648 | 0.00 |
| deviation_theme_stated | -2.757 | 0.00 |
| deviation_set_up | -3.455 | 0.00 |
| deviation_catalyst | -0.103 | 0.00 |
| deviation_debate | -0.263 | 0.00 |
| deviation_b_story | -0.536 | 0.00 |

| | | |
|---|---|---|
| deviation_fun_and_games | -1.374 | 0.00 |
| deviation_break_into_three | -0.640 | 0.00 |
| deviation_finale | -0.902 | 0.00 |

***Table 4: Random Forest Feature Importance***

Only six beats show positive permutation importance scores—a negative score indicates that the model performs better when the beat is not considered. This means beats with a negative score are deemed by the model to be noise. The top three beats ("All Is Lost," "Bad Guys Close In," and "Dark Night Of The Soul" account together for greater than 80% of the total importance in the model. Unlike in the second OLS model, these beats form an identifiable cluster, all occurring consecutively in the second half of the script right after the midpoint. This would suggest that at least within the sample, deviations from Snyder's structure matter most in the second half of the script. However, in order to generalize this observation outside of the sample, we need to have the right summary statistics to back it up.

***Table 5: Summary Statistics of Random Forest Model***

| **Metric** | **Value** |
|---|---|
| Mean Squared Error (MSE) | 474,840.783 |
| Percentage Variance Explained | -23.070 |
| In-sample R-squared ($R^2$) | 0.733 |
| Root Mean Squared Error (RMSE) | 320.781 |

***Table 5: Summary Statistics of Random Forest Model***

There is a fairly strong in-sample fit, as evidenced by the R-squared value of .733. This means that when fitting inside of the dataset, the model is able to attribute 73.3% of variance in global box office to deviations from Snyder's story structure. However, the model fails to perform outside of the sample, with the cross-validated actual percentage variance explained being -23.070. This means that outside of the sample, the model actually performs worse than just predicting the sample mean as a baseline. This, along with a massive root mean squared error of $320.781 million, are telltale signs of overfitting. Therefore, conclusions from this model cannot be generalized outside of the dataset and must be interpreted with extreme caution.

## 6. Conclusion

This thesis aimed to determine if story structure could predict global box office success—specifically, whether adherence to Blake Snyder's *Save the Cat!* beat sheet demonstrated predictive power. Within the broader climate of the entertainment industry, which has increasingly depended upon audience familiarity with known IP from a shrinking pool of professional screenwriters, it is worth investigating whether an optimal formula for narrative structure could help mitigate risk on its own.

All three of my models failed to find evidence that adherence to Snyder's beat structure held predictive power for global box office. The repeated underestimation of high-grossing franchise films like *Avengers: Endgame* and *Star Wars: The Force Awakens* suggests that IP plays a much larger role in financial success than story beats. Future researchers might develop more complex models to account for intellectual property, especially if they also have access to data on star power like actor q-scores.

Further developments in this area would also benefit from looking at marketing budgets and social media virality. Alternatively, it may be worth looking at critical reception as an output instead of financial figures, as story structure might have a greater impact on audience enjoyment than on audience consumption. Nonetheless, these models provided novel insights into the mathematical measurement of story structure that can be adapted into future models, addressing a major gap in existing research.

**Works Cited:**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**Appendix:**

## Table 1: Columns Film–Deviation_Catalyst

| Film | Deviation_Opening Image | Deviation_Theme Stated | Deviation_Set-Up | Deviation_Catalyst |
|---|---|---|---|---|
| A Real Pain | 0.10 | 1.50 | 4.00 | 7.10 |
| A Star Is Born | 0.20 | 10.10 | 2.80 | 8.10 |
| Anchorman | 0.04 | 0.22 | 4.06 | 2.31 |
| Anora | 0.20 | 0.10 | 4.30 | 4.30 |
| Avengers | | | | |
| | 0.22 | 0.39 | 4.32 | 4.05 |
| Endgame | | | | |
| Barbie | 0.10 | 2.80 | 2.40 | 3.10 |
| Before Midnight | 0.02 | 4.27 | 4.12 | 4.01 |
| Black Panther | 0.10 | 2.00 | 4.20 | 0.60 |
| Breakfast Club | 0.40 | 2.00 | 2.50 | 0.50 |
| Bridge Of Spies | 0.02 | 3.76 | 0.41 | 14.79 |
| Call Me By Your | | | | |
| | 0.30 | 4.00 | 3.80 | 12.30 |
| Name | | | | |
| Challengers | 0.20 | 1.00 | 2.80 | 0.10 |
| Chaplin | 0.39 | 3.48 | 3.48 | 4.84 |
| Dazed And | | | | |
| | 0.00 | 1.10 | 0.70 | 0.30 |
| Confused | | | | |
| Django | | | | |
| | 0.90 | 1.50 | 5.00 | 4.30 |
| Unchained | | | | |
| Dumb Money | 0.20 | 0.20 | 0.00 | 1.50 |
| Dunkirk | 1.50 | 0.80 | 2.60 | 2.40 |
| Election | 0.03 | 1.72 | 4.07 | 1.64 |
| Expendables | 0.04 | 0.19 | 4.14 | 11.51 |

| | | | | |
|---|---|---|---|---|
| Fabelmans | 0.21 | 0.33 | 4.31 | 1.87 |
| Fargo | 0.10 | 1.60 | 0.80 | 8.30 |
| Hall Pass | 0.10 | 3.50 | 4.00 | 19.10 |
| Heat | 0.30 | 1.30 | 3.10 | 0.00 |

| Film | Deviation_Opening Image | Deviation_Theme Stated | Deviation_Set-Up | Deviation_Catalyst |
|---|---|---|---|---|
| Hitchhikers Guide | 1.06 | 1.38 | 2.06 | 0.86 |
| How To Train Your Dragon | 0.00 | 7.80 | 4.10 | 1.50 |
| Jojo Rabbit | 0.02 | 1.85 | 4.12 | 2.94 |
| Lady Bird | 0.00 | 2.70 | 4.10 | 5.40 |
| Love Actually | 0.10 | 3.00 | 4.20 | 3.20 |
| Miss Congeniality | 0.10 | 0.60 | 4.00 | 0.10 |
| My Old Ass | 0.07 | 18.80 | 4.03 | 1.72 |
| Napoleon | 0.10 | 1.60 | 0.20 | 2.20 |
| Network | 0.20 | 2.50 | 4.30 | 5.50 |
| No Hard Feelings | 0.02 | 1.69 | 4.12 | 0.60 |
| Oppenheimer | 0.39 | 3.99 | 3.98 | 4.84 |
| Parasite | 0.20 | 1.00 | 2.90 | 3.80 |
| Poltergeist | 0.03 | 4.85 | 4.07 | 2.18 |
| Raiders Of The Lost Ark | 0.00 | 12.80 | 4.10 | 4.50 |
| Red One | 0.10 | 1.70 | 1.90 | 0.90 |
| Shrek | 0.49 | 1.72 | 3.61 | 4.38 |
| Shrek The Third | 0.10 | 5.70 | 2.50 | 1.00 |
| The Departed | 0.90 | 1.90 | 5.00 | 3.70 |
| The Devil Wears Prada | 0.00 | 2.80 | 4.10 | 2.00 |
| The Fault In Our Stars | 0.12 | 1.37 | 2.66 | 2.31 |
| The Favourite | 0.00 | 15.10 | 2.10 | 0.10 |
| The Force | | | | |

| | | | | |
|---|---|---|---|---|
| | 0.10 | 3.00 | 4.20 | 2.90 |
| Awakens | | | | |
| The Other Guys | 0.10 | 0.50 | 4.20 | 1.30 |
| The Princess | 0.10 | 2.10 | 2.60 | 6.10 |

| Film | Deviation_Opening Image | Deviation_Theme Stated | Deviation_Set-Up | Deviation_Catalyst |
|---|---|---|---|---|
| Bride | | | | |
| True Romance | 0.10 | 5.50 | 4.20 | 3.70 |
| Wicked | 0.10 | 2.10 | 1.70 | 7.00 |
| Women Talking | 0.10 | 1.60 | 4.00 | 0.90 |
| Young Frankenstein.Csv | 0.04 | 1.53 | 4.14 | 3.14 |

## Table 2: Columns Deviation_Debate–Deviation_Midpoint

| Deviation_Debate | Deviation_Break | Deviation_B | Deviation_Fun and | |
|---|---|---|---|---|
| | Deviation_Midpoint | Into Two | Story | Games |
| 6.20 | 7.30 | 7.80 | 11.00 | 1.00 |
| 2.90 | 8.70 | 7.80 | 8.00 | 1.80 |
| 1.71 | 1.95 | 1.10 | 9.62 | 6.60 |
| 8.80 | 11.00 | 14.80 | 14.70 | 24.50 |
| 7.21 | 9.69 | 12.13 | 6.71 | 8.22 |
| 7.20 | 7.90 | 5.50 | 2.20 | 12.60 |
| 2.50 | 0.98 | 5.26 | 5.96 | 0.88 |
| 1.20 | 1.90 | 3.90 | 6.10 | 0.80 |
| 3.50 | 6.40 | 10.80 | 16.80 | 13.30 |
| 9.81 | 14.91 | 4.91 | 14.04 | 13.30 |
| 8.80 | 20.00 | 17.90 | 18.80 | 23.20 |

| | | | | |
|---|---|---|---|---|
| 3.60 | 6.20 | 7.40 | 5.80 | 11.50 |
| 0.05 | 1.67 | 0.21 | 0.46 | 2.79 |
| 3.00 | 6.30 | 7.40 | 7.60 | 18.10 |
| 9.00 | 7.60 | 9.10 | 2.90 | 3.60 |
| 5.30 | 9.00 | 11.40 | 11.30 | 24.10 |
| 7.00 | 6.80 | 8.90 | 6.80 | 3.70 |
| 2.91 | 0.48 | 0.58 | 5.19 | 6.48 |

| Deviation_Debate | Deviation_Break Deviation_Midpoint | Deviation_B Into Two | Deviation_Fun and Story | Games |
|---|---|---|---|---|
| 7.34 | 9.20 | 7.28 | 8.79 | 1.72 |
| 1.52 | 4.64 | 4.98 | 5.00 | 10.42 |
| 4.40 | 7.10 | 4.50 | 4.60 | 2.90 |
| 16.20 | 14.30 | 16.80 | 19.00 | 23.00 |
| 4.60 | 3.50 | 4.80 | 3.70 | 9.60 |
| 2.09 | 1.81 | 4.17 | 4.31 | 18.63 |
| 5.50 | 4.80 | 7.70 | 7.70 | 1.90 |
| 4.41 | 5.89 | 4.19 | 2.57 | 1.33 |
| 8.50 | 11.70 | 13.40 | 10.70 | 21.60 |
| 6.80 | 1.90 | 1.30 | 4.60 | 1.50 |
| 5.00 | 6.00 | 3.70 | 3.30 | 19.60 |
| 3.21 | 10.31 | 2.90 | 3.98 | 9.22 |
| 5.30 | 7.30 | 8.00 | 5.00 | 15.40 |
| 8.70 | 9.20 | 11.00 | 9.70 | 20.30 |
| 2.67 | 0.31 | 1.12 | 1.86 | 0.44 |
| 0.05 | 1.67 | 0.21 | 0.46 | 2.79 |
| 8.30 | 10.60 | 11.60 | 13.00 | 12.40 |
| 1.85 | 11.88 | 9.25 | 10.19 | 5.14 |
| 2.40 | 0.40 | 0.70 | 3.70 | 12.50 |

| | | | | |
|---|---|---|---|---|
| 1.80 | 3.00 | 1.20 | 0.10 | 3.50 |
| 0.13 | 6.47 | 13.08 | 18.61 | 9.72 |
| 2.40 | 3.20 | 3.50 | 0.30 | 3.40 |
| 8.20 | 7.60 | 3.50 | 5.00 | 9.90 |
| 0.40 | 2.90 | 3.90 | 1.60 | 6.90 |
| 1.96 | 1.61 | 0.14 | 4.38 | 4.69 |
| 4.10 | 0.80 | 13.00 | 4.50 | 1.00 |
| 1.40 | 0.40 | 0.30 | 0.00 | 1.50 |

| Deviation_Debate | Deviation_Break Into Two | Deviation_B Story | Deviation_Fun and Games | Deviation_Midpoint |
|---|---|---|---|---|
| 6.40 | 5.90 | 4.00 | 2.00 | 10.00 |
| 11.20 | 15.40 | 17.50 | 17.10 | 15.30 |
| 0.90 | 3.50 | 5.90 | 6.20 | 5.40 |
| 2.70 | 1.70 | 1.30 | 9.00 | 6.90 |
| 4.10 | 3.80 | 16.90 | 2.50 | 1.00 |
| 0.42 | 1.15 | 0.48 | 1.90 | 2.59 |

## Table 3: Columns Deviation_Bad Guys Close In–Deviation_Finale

| Deviation_Bad Guys Close In | Deviation_All Is Lost | Deviation_Dark Night of the Soul | Deviation_Break Into Three | Deviation_Finale |
|---|---|---|---|---|
| 5.00 | 2.90 | 3.30 | 4.80 | 4.00 |
| 4.30 | 3.40 | 10.50 | 11.90 | 7.70 |
| 0.43 | 0.77 | 3.72 | 2.05 | 0.85 |
| 31.30 | 28.70 | 26.00 | 11.50 | 4.00 |
| 9.00 | 4.40 | 6.26 | 8.71 | 8.29 |
| 10.30 | 8.10 | 6.60 | 5.90 | 8.50 |
| 4.61 | 0.56 | 3.62 | 16.66 | 9.74 |

| | | | | |
|---|---|---|---|---|
| 4.90 | 4.20 | 4.70 | 3.40 | 5.50 |
| 11.90 | 11.60 | 14.60 | 15.20 | 11.20 |
| 8.89 | 17.22 | 16.29 | 12.71 | 5.83 |
| 15.40 | 16.00 | 12.70 | 9.40 | 5.20 |
| 15.00 | 9.90 | 6.80 | 3.60 | 4.80 |
| 9.76 | 3.47 | 0.40 | 3.51 | 3.78 |
| 23.70 | 19.80 | 2.00 | 0.40 | 1.40 |
| 8.40 | 1.80 | 6.20 | 7.70 | 5.40 |
| 30.20 | 27.80 | 14.40 | 13.20 | 15.90 |
| 9.00 | 2.60 | 0.50 | 4.50 | 0.40 |

| Deviation_Bad Guys Close In | Deviation_All Is Lost | Deviation_Dark Night of the Soul | Deviation_Break Into Three | Deviation_Finale |
|---|---|---|---|---|
| 9.00 | 1.34 | 2.30 | 4.28 | 1.11 |
| 5.55 | 1.72 | 0.29 | 1.34 | 7.41 |
| 18.03 | 9.77 | 9.51 | 9.84 | 12.08 |
| 3.20 | 0.20 | 3.30 | 10.30 | 5.40 |
| 16.00 | 18.90 | 17.30 | 14.80 | 9.00 |
| 14.10 | 9.80 | 4.10 | 2.80 | 2.30 |
| 16.49 | 16.21 | 20.44 | 19.86 | 13.04 |
| 6.20 | 4.50 | 0.90 | 1.70 | 4.80 |
| 6.79 | 3.50 | 2.79 | 2.86 | 0.84 |
| 26.90 | 22.20 | 17.70 | 11.10 | 12.50 |
| 5.20 | 1.10 | 4.20 | 3.60 | 1.20 |
| 18.50 | 16.20 | 13.60 | 12.00 | 9.10 |
| 2.17 | 4.99 | 0.12 | 1.44 | 8.20 |
| 18.60 | 18.10 | 15.00 | 8.90 | 11.90 |
| 22.50 | 15.40 | 13.20 | 7.60 | 1.20 |
| 6.79 | 6.24 | 7.83 | 8.64 | 7.92 |

| | | | | |
|---|---|---|---|---|
| 9.76 | 3.47 | 0.40 | 3.51 | 3.78 |
| 17.90 | 23.40 | 16.00 | 12.70 | 9.10 |
| 1.99 | 1.06 | 1.13 | 2.24 | 1.82 |
| 7.30 | 7.90 | 6.10 | 4.50 | 0.60 |
| 7.80 | 2.70 | 4.20 | 4.00 | 9.40 |
| 11.83 | 9.68 | 7.86 | 10.30 | 6.67 |
| 9.00 | 4.50 | 6.60 | 5.20 | 9.60 |
| 16.90 | 6.90 | 8.20 | 8.10 | 12.00 |
| 10.70 | 3.40 | 2.90 | 4.70 | 4.70 |
| 1.94 | 2.99 | 8.55 | 11.08 | 6.41 |
| 6.00 | 1.50 | 2.80 | 8.10 | 2.30 |

| Deviation_Bad Guys Close In | Deviation_All Is Lost | Deviation_Dark Night of the Soul | Deviation_Break Into Three | Deviation_Finale |
|---|---|---|---|---|
| 5.90 | 4.30 | 1.20 | 2.60 | 5.80 |
| 17.40 | 11.30 | 9.50 | 6.00 | 11.40 |
| 16.30 | 18.90 | 17.10 | 13.50 | 10.00 |
| 0.50 | 4.20 | 1.90 | 0.50 | 5.00 |
| 7.70 | 12.40 | 9.40 | 15.50 | 10.90 |
| 5.10 | 5.40 | 3.80 | 6.10 | 0.30 |
| 2.21 | 2.59 | 4.89 | 4.70 | 5.52 |

## Table 4: Columns Deviation_Final Image–Box_Office_Adj_Mil

| Deviation_Final Image | Average_Deviation | Box_Office_Adj_Mil |
|---|---|---|
| 1.00 | 4.466667 | 7.500000 |
| 0.70 | 5.926667 | 505.975678 |
| 3.77 | 2.613333 | 132.471143 |
| 2.20 | 12.426667 | 59.300000 |
| 7.53 | 6.475333 | 3,178.351648 |

| | | |
|---|---|---|
| 4.30 | 5.833333 | 1,493.988343 |
| 0.88 | 4.271333 | 23.746878 |
| 4.10 | 3.173333 | 1,393.838468 |
| 1.30 | 8.133333 | 53.172356 |
| 2.75 | 9.309333 | 170.977519 |
| 2.40 | 11.346667 | 44.602831 |
| 3.30 | 5.466667 | 99.117402 |
| 2.03 | 2.688000 | 12.389675 |
| 1.70 | 6.233333 | 8.259784 |
| 0.60 | 4.933333 | 439.936719 |
| 1.40 | 11.060000 | 21.372190 |
| 3.70 | 4.080000 | 623.082218 |

| Deviation_Final Image | Average_Deviation | Box_Office_Adj_Mil |
|---|---|---|
| 2.78 | 2.927333 | 25.069199 |
| 5.17 | 4.779333 | 352.725389 |
| 22.22 | 7.648667 | 48.204604 |
| 2.90 | 3.973333 | 44.661786 |
| 0.00 | 12.800000 | 108.736521 |
| 0.60 | 4.306667 | 350.491704 |
| 0.98 | 8.226000 | 144.713959 |
| 0.90 | 4.000000 | 635.933679 |
| 0.88 | 2.998667 | 102.652015 |
| 11.90 | 12.026667 | 98.393881 |
| 2.30 | 2.946667 | 371.397327 |
| 3.90 | 7.713333 | 345.834862 |
| 0.00 | 4.744000 | 5.700000 |
| 1.90 | 7.966667 | 228.589509 |
| 0.70 | 8.800000 | 113.030769 |
| 0.88 | 3.408667 | 90.134888 |
| 2.03 | 2.755333 | 1,007.487094 |
| 8.50 | 10.093333 | 293.080586 |
| 0.93 | 3.907333 | 232.914842 |
| 1.00 | 4.566667 | 1,259.052083 |
| 0.80 | 2.873333 | 185.900000 |
| 1.39 | 7.062667 | 769.085787 |
| 4.20 | 4.080000 | 1,090.395676 |
| 0.70 | 6.566667 | 398.728889 |
| 0.90 | 3.460000 | 449.844444 |
| 0.78 | 3.399333 | 382.074223 |

| | | |
|---|---|---|
| 2.00 | 4.226667 | 110.660430 |

| Deviation_Final Image | Average_Deviation | Box_Office_Adj_Mil |
|---|---|---|
| 1.50 | 2.340000 | 2,587.508000 |
| 1.60 | 6.106667 | 218.702591 |
| 1.60 | 10.986667 | 76.939759 |
| 0.80 | 3.220000 | 24.286624 |
| 0.80 | 5.946667 | 625.400000 |
| 1.00 | 3.773333 | 9.831202 |
| 2.59 | 2.526000 | 89.102415 |